

1 Online appendix for “Convergence-divergence
2 models: Generalized phylogenetic trees modeling
3 gene flow over time” for peer review

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10 Sections, algorithms, definitions, theorems, propositions, figures and expressions intro-
11 duced in the main text are labeled numerically. Any introduced in the online appendix
12 are labeled with a number followed by “A”.

13 **1A Parameter identifiability**

14 Recall that we decompose edges of the principal tree into “diverging sections” and
15 “converging sections”. Converging sections span only a single epoch, while diverging
16 sections may span multiple epochs.

17 Recall that rates and epoch times cannot be identified individually; only their
18 products can be identified. A parameter associated with a diverging section of an

edge than spans a single epoch cannot be identified. Instead, an “average” over the maximum number of epochs a contiguous diverging section can span can be identified. For example, suppose rate matrix $\mathbf{Q}_1$ applies over epoch time t_1 to a diverging section of an edge immediately before an event and rate matrix $\mathbf{Q}_2$ applies over epoch time t_2 to a diverging section of the edge immediately after the event. Then for the 2-state general Markov model, $\exp(\hat{\mathbf{Q}}(t_1 + t_2)) = \exp(\mathbf{Q}_2 t_2) \exp(\mathbf{Q}_1 t_1)$, where $\hat{\mathbf{Q}}$ is again a rate matrix from the 2-state general Markov model. Thus, we apply rate matrix $\hat{\mathbf{Q}}$ to both diverging sections of the edge.

This lack of identifiability result follows from the 2-state general Markov model forming a Lie algebra, sufficient for multiplicative closure of the model class (Sumner et al. 2012a). Suppose α_1 and β_1 and α_2 and β_2 correspond with rate matrices $\mathbf{Q}_1$ and $\mathbf{Q}_2$, respectively. Then by Definition 4, $\frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2}$. It is straightforward to show that if $\hat{\alpha}$ and $\hat{\beta}$ are associated with $\hat{\mathbf{Q}}$, then $\frac{\hat{\alpha}}{\hat{\beta}} = \frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2}$. Thus, the product of the two transition matrices is replaced by a single “average” transition matrix.

All parameters except for those corresponding to the root distribution are of the form $l_i = a_i + b_i = \alpha_i t_i + \beta_i t_i = a_i \left(1 + \frac{b_i}{a_i}\right)$, where i and j are arbitrary parameter indices, $a_i = \alpha_i t_i$ and $b_i = \beta_i t_i$. (Note that these parameters are scalars, whereas $\exp(\hat{\mathbf{Q}}(t_1 + t_2))$ is a matrix.) Since $\frac{\alpha_i}{\beta_i} = \frac{\alpha_j}{\beta_j}$, it follows that $\frac{a_i}{b_i} = \frac{a_j}{b_j}$.

Contiguous diverging sections of an edge — not separated by a converging section — each have a single associated parameter l_i . Furthermore, each convergence group has an associated parameter l_j , in common for all converging sections of edges in the convergence group. In addition to parameters describing the convergence groups and contiguous diverging sections, there is a parameter $\gamma = [\mathbf{\Pi}]_0 - [\mathbf{\Pi}]_1 = \frac{-a_i + b_i}{a_i + b_i}$ describing the difference in probabilities of states 0 and 1 on the root taxon.

To form the set of parameters of a CDM, we consider a particular unique set of diverging and converging sections. Since differences in parameters between contiguous diverging sections cannot be identified, the diverging sections we consider are those

sections on the principal tree between a node or converging section and another node or converging section. Furthermore, since the exact root location on the outgroup edge is not identifiable, we consider one diverging section to be the entire outgroup edge when the principal tree of the CDM is unrooted. The converging sections correspond to individual epochs where there is a convergence group. Converging sections correspond to convergence parameters and diverging sections correspond to divergence parameters.

Note that although this is the general formulation of the parameter space, on a given CDM not all parameters are necessarily identifiable. To obtain an identifiable set of parameters some combinations of the divergence parameters may be required, which we describe in Section 2A.1. For the following sections, the parameters $x_i = \exp(-l_i) \in (0, 1)$ are used for establishing identifiability and distinguishability of CDMs.

2A Identifiability of 4-taxon CDMs

Here we establish whether each 4-taxon CDM is identifiable or not. A 4-taxon CDM is identifiable if there is a one-to-one mapping from the set of generic parameters to the set of realizable phylogenetic tensors (transformed into the Hadamard basis for simplicity). We start by describing the form of the phylogenetic tensors, then proving that, given an arbitrary phylogenetic tensor, a unique parameter set as functions of the phylogenetic tensor exists.

Sumner et al. (2012b) formally describe phylogenetic epoch models in their Definition 6.1 and introduce notation to compute the phylogenetic tensors. We use the same notation for our CDMs.

69 For each 4-taxon CDM, the phylogenetic tensor $\boldsymbol{P}$ is transformed into the
70 Hadamard basis $\hat{\boldsymbol{P}}$ by multiplying by $\boldsymbol{H}_{16} = \boldsymbol{H}_2^{\otimes 4}$, where

$$\boldsymbol{H}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

In the Hadamard basis, the phylogenetic tensor for CDM 5 is

$$\begin{aligned}
 \hat{\mathbf{P}} = \mathbf{H}_{16} \cdot \mathbf{P} = & \begin{bmatrix} q_{0000} \\ q_{0001} \\ q_{0010} \\ q_{0011} \\ q_{0100} \\ q_{0101} \\ q_{0110} \\ q_{0111} \\ q_{1000} \\ q_{1001} \\ q_{1010} \\ q_{1011} \\ q_{1100} \\ q_{1101} \\ q_{1110} \\ q_{1111} \end{bmatrix} \\
 = & \begin{bmatrix} 1 \\ \gamma \\ \gamma \\ \gamma^2 + (1 - \gamma^2) r_{0011} \\ \gamma \\ \gamma^2 + (1 - \gamma^2) r_{0101} \\ \gamma^2 + (1 - \gamma^2) r_{0110} \\ \gamma (\gamma^2 + (1 - \gamma^2) (r_{0011} + r_{0101} + r_{0110} - 2r_{0111})) \\ \gamma \\ \gamma^2 + (1 - \gamma^2) r_{1001} \\ \gamma^2 + (1 - \gamma^2) r_{1010} \\ \gamma (\gamma^2 + (1 - \gamma^2) (r_{0011} + r_{1001} + r_{1010} - 2r_{1011})) \\ \gamma^2 + (1 - \gamma^2) r_{1100} \\ \gamma (\gamma^2 + (1 - \gamma^2) (r_{0101} + r_{1001} + r_{1100} - 2r_{1101})) \\ \gamma (\gamma^2 + (1 - \gamma^2) (r_{0110} + r_{1010} + r_{1100} - 2r_{1110})) \\ \gamma^2 (\gamma^2 + (1 - \gamma^2) (r_{0011} + r_{0101} + r_{0110} + r_{1001} + r_{1010} + r_{1100} \\ - 2(r_{0111} + r_{1011} + r_{1101} + r_{1110} - 2\delta))) + (1 - \gamma^2)^2 r_{1111} \end{bmatrix}. \tag{1A}
 \end{aligned}$$

We see immediately that $\gamma = q_{0001} = q_{0010} = q_{0100} = q_{1000}$. Thus, we can express $r_{0011}, r_{0101}, \dots, r_{1111}$ and δ as functions of the phylogenetic tensor elements and γ . Finally, we can express the parameters of the CDMs as functions of $r_{0011}, r_{0101}, \dots, r_{1111}$ and δ . For this final step, we use algebraic geometry, including ideals and their Gröbner bases.

See Mathematica file S2.nb (text version S3.txt) on <https://github.com/jonathanmitchell88/CDMsSI> for a derivation of Equation (1A) and equations for $r_{0011}, r_{0101}, \dots, r_{1111}$ and δ in terms of x_i and y_i (products of x_i) for CDM 5. CDMs 1 – 4 are all nested in CDM 5. Thus, their phylogenetic tensors are also in the form of Equation (1A). However, the equations for $r_{0011}, r_{0101}, \dots, r_{1111}$ and δ involve different expressions of x_i and y_i .

For the proof that follows, the order of parameters is as in Figure 2, with $x_i = \exp(-(a_i + b_i)) \in (0, 1)$ for all $i \in \{1, 2, \dots, 11\}$. Note again that the exact location of the root on the outgroup edge is not identifiable; t_1 corresponds to the sum of epoch times of epochs from the root to the outgroup added to the first epoch time below the root.

To establish whether a CDM is identifiable or not, we must first determine a maximal set of independent elements of the transformed phylogenetic tensor. That is, a set with maximum cardinality such that there are no algebraic equations — equalities — involving multiple elements of the set. If the cardinality of the set equals the number of parameters, then the CDM is identifiable. For example, we can see that invariants $q_{0001} = q_{0010} = q_{0100} = q_{1000} = \gamma$ are equalities on all CDMs. Thus, we can only include one of $q_{0001}, q_{0010}, q_{0100}$ and q_{1000} in the set.

To determine all such equalities, for a given CDM with $l + 1$ parameters $x_1, x_2, \dots, x_l, \gamma$, we construct the ideal,

$$I = \langle r_{0011} - f_{0011}(x_1, x_2, \dots, x_l), r_{0101} - f_{0101}(x_1, x_2, \dots, x_l), \dots, \rangle$$

$$r_{1111} - f_{1111}(x_1, x_2, \dots, x_l), \delta - f_\delta(x_1, x_2, \dots, x_l)\rangle$$

$$\subseteq \mathbb{Q}[x_1, x_2, \dots, x_l, r_{0011}, r_{0101}, \dots, r_{1111}, \delta],$$

97 where each $r_{ijkl} - f_{ijkl}(x_1, x_2, \dots, x_l)$ and $\delta - f_\delta(x_1, x_2, \dots, x_l)$ is identically zero.
 98 (We can ignore γ since $q_{ijkl} = \gamma^2 + (1 - \gamma^2)r_{ijkl}$ and including any of these invariants
 99 does not help us to determine any invariants involving multiple variables $r_{0011}, r_{0101},$
 100 $\dots, r_{1111}, \delta$.)

101 In the Macaulay2 file S4.m2 (output file S5.txt) on [https://github.com/](https://github.com/jonathanmitchell88/CDMsSI)
 102 [jonathanmitchell88/CDMsSI](https://github.com/jonathanmitchell88/CDMsSI) we derive the (reduced) Gröbner basis for this ideal for
 103 a particular monomial order for CDM 5. Below we outline how this Gröbner basis is
 104 computed.

105 In the Mathematica file S2.nb (text version S3.txt) we derive the following
 106 equations to input into the generators of the ideal:

$$\left\{ \begin{array}{l} f_{0011} = x_4 x_5 x_6 x_7 x_9 x_{10}, \\ f_{0101} = x_{10} x_{11} (1 - x_9 (1 - x_2 x_3 x_4 x_6 x_8)), \\ f_{0110} = x_7 x_8 x_9 x_{11} (1 - x_6 (1 - x_2 x_3 x_5)), \\ f_{0111} = x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_{10} x_{11}, \\ f_{1001} = x_1 x_2 x_4 x_9 x_{10}, \\ f_{1010} = x_1 x_2 x_5 x_6 x_7, \\ f_{1011} = x_1 x_2 x_4 x_5 x_6 x_7 x_9 x_{10}, \\ f_{1100} = x_1 x_3 x_6 x_8 x_9 x_{11}, \\ f_{1101} = x_1 x_2 x_3 x_4 x_6 x_8 x_9 x_{10} x_{11}, \\ f_{1110} = x_1 x_2 x_3 x_5 x_6 x_7 x_8 x_9 x_{11}, \\ f_{1111} = x_1 x_7 x_{10} x_{11} (x_4 x_8 x_9 (x_2 (1 - x_6) + x_3 x_5 x_6) + x_2 x_5 x_6 (1 - x_9)), \\ f_{\delta} = x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_{10} x_{11}. \end{array} \right. \quad (2A)$$

107 The functions $f_{0011} = f_{0011}(x_1, x_2, \dots, x_l)$, $f_{0101} = f_{0101}(x_1, x_2, \dots, x_l)$, ...,
 108 $f_{1111} = f_{1111}(x_1, x_2, \dots, x_l)$ and $f_{\delta} = f_{\delta}(x_1, x_2, \dots, x_l)$ depend on the CDM in
 109 question, for example, CDM 5.

110 The monomial order is the elimination order, eliminating the block $x_1, x_2, \dots, x_l$,
 111 with graded reverse lexicographic order on each block, $x_1 > x_2 > \dots > x_l$ and
 112 $r_{0011} > r_{0101} > \dots > r_{1111} > \delta$.

113 Next, we compute the (reduced) Gröbner basis I_G of I . Then $I_{G,q} = I_G \cap$
 114 $\mathbb{R}[r_{0011}, r_{0101}, \dots, r_{1111}, \delta]$ is a Gröbner basis for the elimination ideal involving only
 115 $r_{0011}, r_{0101}, \dots, r_{1111}, \delta$.

116 Note that q_{1111} is a function of both r_{1111} and δ , the only element of $\hat{\mathbf{P}}$ that is
 117 a function of either. Thus, the maximum cardinality set can include at most one of
 118 r_{1111} and δ . In S4.m2 we find that when eliminating r_{1111} there are no generators
 119 that involve δ . Thus, r_{1111} is eliminated and δ is another independent variable of the
 120 system when r_{1111} is eliminated.

121 Note that there are still some algebraic equations — equalities — involving some
 122 elements of $\{r_{0011}, r_{0101}, \dots, r_{1110}, \delta\}$. In S4.m2 (output file S5.txt) we find the largest
 123 cardinality subset with no algebraic equations involving multiple elements. This car-
 124 dinality, plus one for γ , is the degrees of freedom of the phylogenetic tensor. Given a
 125 set of parameters of the CDM, if this degrees of freedom is less than the number of
 126 parameters, then the system of polynomial equations is underdetermined and that set
 127 of parameters is not identifiable. (Note that some individual parameters may still be
 128 identifiable.) Otherwise, the set of parameters is identifiable. If that set of param-
 129 eters is not identifiable, it may be possible to combine the parameters in a such a way
 130 that the new set of parameters is identifiable.

131 2A.1 Proof of Proposition 1

132 See S4.m2 (output file S5.txt) and S6.m2 (output file S7.txt) on [https://github.com/](https://github.com/jonathanmitchell88/CDMsSI)
 133 [jonathanmitchell88/CDMsSI](https://github.com/jonathanmitchell88/CDMsSI) for the computations of the (reduced) Gröbner bases of
 134 the ideals in this proof.

135 **Proposition 1** *CDM 5, with parameter set $\{\gamma, y_1, y_2, \dots, y_9\}$, is identifiable.*

136 *Proof* In S4.m2 (output file S5.txt), we see that there are 9 elements of
 137 $\{r_{0011}, r_{0101}, \dots, r_{1110}, \delta\}$ that are free to vary. However, CDM 5 has 11 parameters exclud-
 138 ing γ . Thus, this set of parameters is not identifiable. However, recall in Section 1A that
 139 taking some products of x_i parameters may be required to obtain a set of identifiable

140 parameters. Since there are 9 elements of $\{r_{0011}, r_{0101}, \dots, r_{1110}, \delta\}$ that are free to vary, we
 141 desire a set of 9 parameters.

142 In S2.nb (text version S3.txt), we express $f_{0011}, f_{0101}, \dots, f_{1111}, \delta$ in terms of the set of
 143 parameters $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$. Precisely,

$$\left\{ \begin{array}{l} y_1 = x_1, \\ y_2 = x_2, \\ y_3 = x_3 x_8 x_{11}, \\ y_4 = x_4 x_{10}, \\ y_5 = x_5 x_7, \\ y_6 = x_6, \\ y_7 = x_7 x_8 x_{11}, \\ y_8 = x_9, \\ y_9 = x_{10} x_{11}. \end{array} \right.$$

144 In S6.m2 (output file S7.txt), we see that this set of parameters is identifiable. We note
 145 that $x_i \in (0, 1)$ for all $i \in \{1, 2, \dots, 11\}$. It follows that $r_{0011}, r_{0101}, \dots, r_{1111}, \delta \in (0, 1)$ and
 146 $y_i \in (0, 1)$ for all $i \in \{1, 2, \dots, 9\}$. In S2.nb (text version S3.txt), we see that the solutions to

the system are

$$\left\{ \begin{array}{l} y_1 = \frac{\delta}{r_{0111}}, \\ y_2 = \frac{r_{0111}\sqrt{r_{1001}r_{1010}}}{\delta\sqrt{r_{0011}}}, \\ y_3 = \frac{\delta}{\sqrt{r_{0011}r_{1001}r_{1010}}}, \\ y_4 = \frac{r_{1101}\delta\sqrt{r_{0011}}}{r_{0111}r_{1100}\sqrt{r_{1001}r_{1010}}}, \\ y_5 = \frac{\delta}{r_{1101}}, \\ y_6 = \frac{r_{1101}\sqrt{r_{0011}r_{1010}}}{\delta\sqrt{r_{1001}}}, \\ y_7 = \frac{\delta(r_{0110}r_{1101}\delta\sqrt{r_{0011}} - r_{0111}^2r_{1100}\sqrt{r_{1001}r_{1010}})}{r_{0111}r_{1100}\sqrt{r_{0011}r_{1001}}(\delta\sqrt{r_{1001}} - r_{1101}\sqrt{r_{0011}r_{1010}})}, \\ y_8 = \frac{r_{0111}r_{1001}r_{1100}}{r_{1101}\delta}, \\ y_9 = \frac{r_{1101}(r_{0101}\delta - r_{0111}r_{1101})}{r_{1101}\delta - r_{0111}r_{1001}r_{1100}}. \end{array} \right. \quad (3A)$$

Thus, the parameter set $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9, \gamma\}$ on CDM 5 is identifiable.

□

Since CDMs 1 – 4 are all nested in CDM 5, the transformed phylogenetic tensors of CDMs 1 – 4 can be determined directly from that of CDM 5 by setting some parameters x_i to 1. Similarly, it is straightforward to prove that the equivalent sets of y_i parameters are identifiable for each of CDMs 1 – 4. The numbers of degrees of freedom for the phylogenetic tensors of CDMs 1 – 5 are 6, 7, 8, 9 and 10, respectively.

3A Proof of Theorem 2

Recall from Definition 5 that if all points in the intersection of the sets of possible realizable phylogenetic tensors correspond to non-generic parameters in at least one of the parameters spaces of the CDMs, then the CDMs are distinguishable. We provide a technical justification for this definition here.

Recall that CDMs $\mathcal{N}_1$ and $\mathcal{N}_2$ have sets of possible realizable phylogenetic tensors $\mathcal{P}_1$ and $\mathcal{P}_2$. Then $\mathcal{P}_1$ is defined by a system of polynomial equations

(phylogenetic invariants) in the form $g_i = 0$ and inequalities (parameter constraints). Then suppose we have affine variety $\mathcal{V}_1 = \mathcal{V}_1(g_1, g_2, \dots, g_s) = \{(a_1, a_2, \dots, a_r) \in \mathbb{Q}^r \mid g_i(a_1, a_2, \dots, a_r) = 0 \text{ for all } i \in \{1, 2, \dots, s\}\}$.

Recall that Θ'_1 corresponds to parameters in Θ_1 corresponding to realizable phylogenetic tensors in $\mathcal{P}_1 \cap \mathcal{P}_2$. Suppose Θ'_1 corresponds to non-generic parameters in Θ_1 . Suppose also that $\mathcal{V}'_1$ is the affine variety defined by the system of polynomials for $\mathcal{P}_1$ restricted to $\mathcal{P}_1 \cap \mathcal{P}_2$. Then $\mathcal{V}'_1 = \mathcal{V}_1 \cap H_1 \cap H_2 \cap \dots \cap H_w$, where $H_1, H_2, \dots, H_w$ are affine varieties defined by hypersurfaces (one polynomial equation each). By Exercise 1.8 of Chapter 1 of [Hartshorne \(2013\)](#), every irreducible component of $\mathcal{V}'_1$ has dimension $\dim(\mathcal{V}_1) - w$. By Corollary 9 of Chapter 9, Section 4 of [Cox et al. \(1997\)](#), $\dim(\mathcal{V}'_1)$ is the largest of the dimensions of its irreducible components. Thus, $\dim(\mathcal{V}'_1) = \dim(\mathcal{V}_1) - w$. Thus, the set of points in $\mathcal{P}_1 \cap \mathcal{P}_2$ is “small” compared to the set of points in $\mathcal{P}_1$.

For a more in-depth proof of Theorem 2, we could consider the (reduced) Gröbner bases of the ideals representing the parameter spaces of the CDMs and show that each CDM has a unique Gröbner basis. The Gröbner basis for CDM 5 has already been computed in Section 2A. However, computation of the Gröbner bases is slow and some bases contain many generators. Instead, it is sufficient to consider only a few constraints for each parameter space that exist for some CDMs and not others, greatly simplifying the proof.

To prove distinguishability, we show that only non-generic (or possibly no) parameters of one CDM correspond to points in the intersection of possible realizable phylogenetic tensors for two CDMs. To do this, we could show that there are polynomial equations (phylogenetic invariants) involving the elements of the transformed phylogenetic tensor $(q_{0000}, q_{0001}, \dots, q_{1111})$ that exist for one CDM, but not the other. However, noting that $\gamma = q_{0001} = q_{0010} = q_{0100} = q_{1000}$ for all our 4-taxon CDMs, it

188 is easier to find polynomial equations involving the elements $r_{0000}, r_{0001}, \dots, r_{1111}$ and
 189 δ and use these to establish distinguishability.

190 **Theorem 2** *All pairs of 4-taxon leaf-labeled CDMs of Section 4.2 are distinguishable.*

191 *Proof* Suppose first that one CDM is nested in the other. With no loss of generality, assume
 192 CDM $\mathcal{N}_2$ with parameter space Θ_2 and set of possible realizable phylogenetic tensors $\mathcal{P}_2$
 193 is nested in CDM $\mathcal{N}_1$ with parameter space Θ_1 and set of possible realizable phylogenetic
 194 tensors $\mathcal{P}_1$. Then Θ_2 can be obtained from Θ_1 by fixing some parameter(s). However, since
 195 Θ_2 corresponds to fixing some parameter(s) to be on the boundary of Θ_1 (setting edge lengths
 196 or epochs to zero) and is thus in the closure of Θ_1 but not Θ_1 itself, by Assumption 12,
 197 $\Theta_1 \cap \Theta_2 = \emptyset$. Thus, by the identifiability of all CDMs (see Proposition 1 for CDM 5),
 198 $\mathcal{P}_1 \cap \mathcal{P}_2 = \emptyset$. (Note that identifiability of $\mathcal{N}_1$ still holds if closure points are included in Θ_1 .)

199 All that is left to prove is that any two CDMs that differ only in their leaf labelings are
 200 distinguishable. It then follows that two CDMs that differ in their leaf labelings and also
 201 their numbers of parameters are distinguishable. To see this, suppose $\mathcal{N}_1$ and $\mathcal{N}_2$ are distin-
 202 guishable, with corresponding affine varieties $\mathcal{V}_1$ and $\mathcal{V}_2$ and corresponding sets of possible
 203 realizable phylogenetic tensors $\mathcal{P}_1$ and $\mathcal{P}_2$. Then $\mathcal{P}_1 \cap \mathcal{P}_2$ corresponds to a some lower dimen-
 204 sional subvariety of both $\mathcal{V}_1$ and $\mathcal{V}_2$. Then if $\mathcal{N}'_1$ is nested in $\mathcal{N}_1$, $\mathcal{P}'_1 \cap \mathcal{P}_2$ must also correspond
 205 to some lower dimensional subvariety of $\mathcal{V}_2$.

206 The notation that follows is consistent with that of Section 2A. Recall that $y_i \in (0, 1)$ for
 207 all $i \in \{1, 2, \dots, 9\}$.

208 CDM 5

209 See S8.nb (text version S9.txt) on <https://github.com/jonathanmitchell88/CDMsSI> for
 210 proofs of the following claims.

211 For leaf labelings $(o, (a, (b, c)))$ and $(o, (a, (c, b)))$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 = 0,$$

212 while for the other leaf labelings

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 > 0.$$

213 Thus, if $\mathcal{P}_1$ is the set of realizable phylogenetic tensors corresponding to either leaf labeling
 214 $(o, (a, (b, c)))$ or $(o, (a, (c, b)))$ and $\mathcal{P}_2$ is the set of realizable phylogenetic tensors correspond-
 215 ing to a different leaf labeling, then $\mathcal{P}_1 \cap \mathcal{P}_2 = \emptyset$. (Note that the choice of taxon a as the
 216 sister taxon to the outgroup when the 4-taxon CDM is unrooted is arbitrary.)

217 Thus, we need only show that CDMs with leaf labelings $(o, (a, (b, c)))$ and $(o, (a, (c, b)))$ are
 218 distinguishable. Let $\mathcal{P}_1$ and $\mathcal{P}_3$ be the sets of realizable phylogenetic tensors corresponding to
 219 leaf labelings $(o, (a, (b, c)))$ and $(o, (a, (c, b)))$. We show that all points in $\mathcal{P}_1 \cap \mathcal{P}_3$ correspond
 220 to non-generic points in either Θ_1 or Θ_3 . Letting y_i be the parameters corresponding to leaf
 221 labeling $(o, (a, (b, c)))$ and z_i corresponding to $(o, (a, (c, b)))$, we equate the equations for each
 222 element of the two phylogenetic tensors and solve for the z_i parameters,

$$\left\{ \begin{array}{l} y_4 y_5 y_6 y_8 = z_4 z_5 z_6 z_8, \\ y_9 (1 - y_8) + y_2 y_3 y_4 y_6 y_8 = z_8 (z_7 (1 - z_6) + z_2 z_3 z_5 z_6), \\ y_8 (y_7 (1 - y_6) + y_2 y_3 y_5 y_6) = z_9 (1 - z_8) + z_2 z_3 z_4 z_6 z_8, \\ y_2 y_3 y_4 y_5 y_6 y_8 = z_2 z_3 z_4 z_5 z_6 z_8, \\ y_1 y_2 y_4 y_8 = z_1 z_2 z_5 z_6, \\ y_1 y_2 y_5 y_6 = z_1 z_2 z_4 z_8, \\ y_1 y_2 y_4 y_5 y_6 y_8 = z_1 z_2 z_4 z_5 z_6 z_8, \\ y_1 y_3 y_6 y_8 = z_1 z_3 z_6 z_8, \\ y_1 y_2 y_3 y_4 y_6 y_8 = z_1 z_2 z_3 z_5 z_6 z_8, \\ y_1 y_2 y_3 y_5 y_6 y_8 = z_1 z_2 z_3 z_4 z_6 z_8, \\ y_1 (y_4 y_8 (y_2 y_7 (1 - y_6) + y_3 y_5 y_6) \\ \quad + y_2 y_5 y_6 y_9 (1 - y_8)) = z_1 (z_4 z_8 (z_2 z_7 (1 - z_6) + z_3 z_5 z_6) \\ \quad + z_2 z_5 z_6 z_9 (1 - z_8)), \\ y_1 y_2 y_3 y_4 y_5 y_6 y_8 = z_1 z_2 z_3 z_4 z_5 z_6 z_8. \end{array} \right.$$

223 Solving this system of equations — see S10.m2 (output file S11.txt) and the expressions
 224 simplified in S8.nb (text version S9.txt) on <https://github.com/jonathanmitchell88/CDMsSI>
 225 — we obtain

$$z_1 z_2^2 z_3 z_4 z_6 z_8 (1 - z_6) (1 - z_8) (z_4 z_7 z_8 - z_5 z_9) = 0.$$

226 Since $z_i \in (0, 1)$ for all $i \in \{1, 2, \dots, 8\}$, we must have $z_4 z_7 z_8 - z_5 z_9 = 0$. These points are
 227 non-generic points in the parameter space. Thus, for CDM 5, any two CDMs with different
 228 leaf labelings are distinguishable.

229 CDM 4

230 The proof is identical to that of CDM 5, but with the addition of $y_9 = z_9 = 1$. Again, see
 231 S8.nb (text version S9.txt) and S10.m2 (output file S11.txt). We obtain

$$z_1 z_2 z_3 z_4 z_5 z_6 z_8 (1 - z_6) (1 - z_7 z_8) = 0,$$

232 which has no solutions since $z_i \in (0, 1)$ for all $i \in \{1, 2, \dots, 8\}$. Thus, for CDM 4, $\mathcal{P}_1 \cap \mathcal{P}_3 = \emptyset$
 233 and any two CDMs with different leaf labelings are distinguishable.

234 CDM 3

235 See S8.nb (text version S9.txt) for proofs of the following claims.

236 For leaf labeling pairs $(o, (a, (b, c)))$ and $(o, (c, (b, a)))$, $(o, (a, (c, b)))$ and $(o, (b, (c, a)))$ and
 237 $(o, (b, (a, c)))$ and $(o, (c, (a, b)))$,

$$\begin{cases} r_{0101} r_{1010} < r_{0011} r_{1100}, r_{0110} r_{1001}, \\ r_{0110} r_{1001} < r_{0011} r_{1100}, r_{0101} r_{1010}, \\ r_{0011} r_{1100} < r_{0101} r_{1010}, r_{0110} r_{1001}, \end{cases}$$

238 respectively, where each equation corresponds to a leaf labeling pair. Thus, any CDM from
 239 one pair is distinguishable from a CDM from another pair.

240 All that is left is to prove that CDMs from an arbitrary pair, for example, $(o, (a, (b, c)))$
 241 and $(o, (c, (b, a)))$, are distinguishable. For leaf labeling $(o, (a, (b, c)))$, but not $(o, (c, (b, a)))$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 = 0.$$

242 For leaf labeling $(o, (c, (b, a)))$, but not $(o, (a, (b, c)))$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 > 0.$$

243 Thus for CDM 3, any two CDMs with different leaf labelings are distinguishable.

244 CDM 2

245 See S8.nb (text version S9.txt) for proofs of the following claims.

246 The constraints for CDM 2 include those described above for CDM 3. Thus for CDM 2,
 247 any two CDMs with different leaf labelings are distinguishable.

248 CDM 1

249 See S8.nb (text version S9.txt) for proofs of the following claims.

250 For leaf labelings $(o, (a, (b, c)))$, $(o, (b, (a, c)))$ and $(o, (c, (a, b)))$,

$$\begin{cases} r_{0101}r_{1010} = r_{0110}r_{1001} < r_{0011}r_{1100}, \\ r_{0011}r_{1100} = r_{0110}r_{1001} < r_{0101}r_{1010}, \\ r_{0011}r_{1100} = r_{0101}r_{1010} < r_{0110}r_{1001}, \end{cases}$$

251 respectively. Thus, for CDM 1 any two CDMs with different leaf labelings are distinguishable.

252 □

253 4A Proof of Theorem 3

254 **Theorem 3** (Distance on the topology of an N -taxon principal tree) *Let $\mathcal{T}$ be a principal*
 255 *tree, with outgroup o . Suppose $\mathcal{T}$ is given the rooted triple metrization. Then the distance*

256 $d_{\mathcal{T}}(x, y)$ between leaf taxa x and y is

$$d_{\mathcal{T}}(x, y) = \begin{cases} 0 & \text{if } x=y, \\ 2N-2 & \text{if } x \neq y \text{ and one of } x=o, y=o, \\ 2|R_{x,y}|+2 & \text{otherwise,} \end{cases}$$

257 where $R_{x,y}$ is the set of rooted 4-taxon principal trees displayed on $\mathcal{T}$ with outgroup o
 258 displaying both x and y , where x and y are non-sisters.

259 *Proof* Clearly, if $x = y$ then $d_{\mathcal{T}}(x, y) = 0$.

260 Next suppose $x \neq y$ and one of $x = o, y = o$. With no loss of generality, assume $y = o$.

261 Then

$$d_{\mathcal{T}}(x, y) = d_{\mathcal{T}}(x, o) = d_{\mathcal{T}}(x, v) + d_{\mathcal{T}}(v, o),$$

262 where v is the most recent common ancestor (MRCA) of x and o . Since $y = o$, v must be the
 263 root of $\mathcal{T}$. Then from the rooted triple metrization, by the same arguments as [Rhodes \(2019\)](#),

$$d_{\mathcal{T}}(x, v) = d_{\mathcal{T}}(v, o) = N - 1$$

264 and

$$d_{\mathcal{T}}(x, o) = 2N - 2.$$

265 Finally, suppose $x \neq y$ and $x, y \neq o$. Again suppose that v is the MRCA of x and y . Then
 266 again by the same arguments as [Rhodes \(2019\)](#),

$$d_{\mathcal{T}}(x, y) = 2k - 2,$$

267 where k is the number of leaf taxa descended from v .

268 For x and y to be non-sisters on a rooted 4-taxon principal tree displayed on $\mathcal{T}$ with
 269 outgroup o , we require the leaf taxon that is not x, y or o to be one of the $k - 2$ leaf taxa
 270 descended from v that is not x or y . Thus,

$$|R_{x,y}| = k - 2$$

271 and

$$d_{\mathcal{T}}(x, y) = 2 |R_{x,y}| + 2.$$

272

□

273 5A Inferring topologies of N -taxon principal trees

274 We prove that consistent inference of the topology of the N -taxon principal tree fol-
275 lows from consistent inference of the principal trees of the displayed 4-taxon CDMs.
276 However, it is possible that a displayed 4-taxon CDM does not meet the assumptions
277 of Section 3.2. Specifically, even if an N -taxon CDM meets the assumptions, some dis-
278 played 4-taxon CDMs may have sister convergence. By assuming that all convergence
279 parameters of the N -taxon CDM are sufficiently “small”, then all convergence param-
280 eters of the displayed 4-taxon CDMs, including those of sister convergence groups of
281 the displayed 4-taxon CDMs are “small”. Then all topologies of the displayed 4-taxon
282 principal trees are inferred consistently by Algorithm 1.

283 To prove this result, we first prove a proposition similar to Proposition 1.2 of
284 [Haughton \(1988\)](#). Proposition 1.2 states that if the generating model is among the
285 set of candidate models, the probability that the model selected by the BIC is the
286 generating model converges to 1. Our adaptation relaxes Proposition 1.2, such that
287 none of the candidate models are the generating model, but some candidate models
288 are sufficiently “close” to the generating model. That is, the generating parameter is
289 a “small” perturbation from a point in the parameter space of a candidate model. We
290 then use our proposition to prove that all topologies of the displayed 4-taxon principal
291 trees are inferred consistently by Algorithm 1.

292 For the following proposition, $f(X, \phi) = \exp(X\phi - b(\phi))$ is the density for a
293 regular exponential family, m_1 and m_2 are the natural parameter spaces of two models,
294 $\text{int } \Theta$ is the interior of some topological space Θ , $\overline{m}_1$ and $\overline{m}_2$ are the Zariski closures of

295 m_1 and m_2 , respectively and $E_\theta X_i = \nabla b(\theta)$ is the expected value of random variable
 296 X_i given generating parameter θ . The natural parameter space of the exponential
 297 family is as defined in [Lehmann and Romano \(2005\)](#), page 51. Note that for our
 298 multinomially distributed random variables, the natural parameter space is the set of
 299 realizable phylogenetic tensors. The function $g(\phi) = \nabla b(\theta) \phi - b(\phi)$ for $\phi \in \Theta$ attains
 300 its unique maximum at θ ([Barndorff-Nielsen 1978](#)).

301 **Proposition 2A** *Let m_1 and m_2 be two different models satisfying $m_1 \cap m_2 = \emptyset$. Then there*
 302 *exists some $\theta \in \text{int } \Theta$, $\theta \notin \overline{m}_1, \theta \notin \overline{m}_2$, with a neighborhood $\mathfrak{N}$ of θ such that $\mathfrak{N} \cap m_1 = \emptyset$,*
 303 *$\mathfrak{N} \cap m_2 \neq \emptyset$ and*

$$\lim_{n \rightarrow \infty} P_\theta^n(\gamma(n, 1) < \gamma(n, 2)) = 1.$$

304 *Proof* The proof requires only a slight modification to the proof of Proposition 1.2 of
 305 [Haughton \(1988\)](#).

306 From [Haughton \(1988\)](#), since $\mathfrak{N} \cap m_1 = \emptyset$,

$$\sup_{\phi \in m_1 \cap \Theta} \nabla b(\theta) \phi - b(\phi) + \epsilon \leq \nabla b(\theta) \theta - b(\theta) \quad (4A)$$

307 and asymptotically with probability 1,

$$\left| \sup_{\phi \in m_i \cap \Theta} (Y_n \phi - b(\phi)) - \sup_{\phi \in m_i \cap \Theta} \nabla b(\theta) \phi - b(\phi) \right| < \frac{\epsilon}{4}, \quad (5A)$$

308 where $\epsilon > 0$.

309 Although $\mathfrak{N} \cap m_2 \neq \emptyset$, $g(\phi)$ attains its maximum at θ and $\theta \notin \overline{m}_2$. Thus, we can choose
 310 $\tilde{\epsilon} > 0$ such that

$$\sup_{\phi \in m_2 \cap \Theta} \nabla b(\theta) \phi - b(\phi) + \tilde{\epsilon} = \nabla b(\theta) \theta - b(\theta). \quad (6A)$$

311 We consider the two possible signs of the argument of the absolute value in Inequal-
 312 ity (5A). If

$$\sup_{\phi \in m_i \cap \Theta} (Y_n \phi - b(\phi)) - \sup_{\phi \in m_i \cap \Theta} \nabla b(\theta) \phi - b(\phi) \geq 0,$$

313 then from Inequality 5A,

$$\sup_{\phi \in m_1 \cap \Theta} (Y_n \phi - b(\phi)) < \sup_{\phi \in m_1 \cap \Theta} \nabla b(\theta) \phi - b(\phi) + \frac{\epsilon}{4}.$$

314 Similarly, if

$$\sup_{\phi \in m_i \cap \Theta} (Y_n \phi - b(\phi)) - \sup_{\phi \in m_i \cap \Theta} \nabla b(\theta) \phi - b(\phi) < 0,$$

315 then

$$\begin{aligned} \sup_{\phi \in m_1 \cap \Theta} (Y_n \phi - b(\phi)) &< \sup_{\phi \in m_1 \cap \Theta} \nabla b(\theta) \phi - b(\phi) \\ &< \sup_{\phi \in m_1 \cap \Theta} \nabla b(\theta) \phi - b(\phi) + \frac{\epsilon}{4}. \end{aligned}$$

316 Thus, from Inequalities (4A) and (5A), asymptotically with probability 1,

$$\sup_{\phi \in m_1 \cap \Theta} (Y_n \phi - b(\phi)) < \sup_{\phi \in m_1 \cap \Theta} \nabla b(\theta) \phi - b(\phi) + \frac{\epsilon}{4} \leq \nabla b(\theta) \theta - b(\theta) - \frac{3\epsilon}{4}. \quad (7A)$$

317 By similar arguments, from Inequality (5A) and Equation (6A), asymptotically with
318 probability 1,

$$\sup_{\phi \in m_2 \cap \Theta} (Y_n \phi - b(\phi)) > \sup_{\phi \in m_2 \cap \Theta} \nabla b(\theta) \phi - b(\phi) - \frac{\epsilon}{4} = \nabla b(\theta) \theta - b(\theta) - \tilde{\epsilon} - \frac{\epsilon}{4}. \quad (8A)$$

319 By Inequalities (7A) and (8A),

$$\begin{aligned} \sup_{\phi \in m_1 \cap \Theta} (Y_n \phi - b(\phi)) &< \nabla b(\theta) \theta - b(\theta) - \frac{3\epsilon}{4} \\ &= \nabla b(\theta) \theta - b(\theta) - \tilde{\epsilon} - \frac{\epsilon}{4} + \tilde{\epsilon} - \frac{\epsilon}{2} \\ &< \sup_{\phi \in m_2 \cap \Theta} (Y_n \phi - b(\phi)) + \tilde{\epsilon} - \frac{\epsilon}{2} \\ &= \sup_{\phi \in m_2 \cap \Theta} (Y_n \phi - b(\phi)) - \delta, \end{aligned}$$

320 where $\delta = \frac{\epsilon}{2} - \tilde{\epsilon}$.

321 If it is possible to choose $\delta > 0$, then asymptotically with probability 1,

$$\sup_{\phi \in m_1 \cap \Theta} (Y_n \phi - b(\phi)) + \delta < \sup_{\phi \in m_2 \cap \Theta} (Y_n \phi - b(\phi)).$$

322 We are free to choose any $\theta \in \text{int } \Theta$. Thus, we choose θ to be an arbitrarily small pertur-
 323 bation from some point in m_2 . Then $\tilde{\epsilon} > 0$ is arbitrarily small and $\delta > 0$. The remainder of
 324 the proof then follows from [Haughton \(1988\)](#).

325 □

326 A convergence group on the generating N -taxon CDM may be a sister convergence
 327 group on some displayed 4-taxon CDMs and a non-sister convergence group on others.
 328 Thus, we must assume that all convergence parameters of the generating N -taxon
 329 CDM are “small” relative to the divergence parameters.

330 Next, we adapt Theorem 3 of [Steel \(1992\)](#) to prove that the N -taxon principal tree
 331 can be identified from the set of 4-taxon principal trees that include the outgroup.

332 **Theorem 3A** (Steel, 1992) *For a set of rooted triples R , $\langle R \rangle = \{T\}$ if and only if R is*
 333 *consistent with T , and for each internal edge e of T there is a rooted triple in R which*
 334 *distinguishes e .*

335 The consequence of Theorem 3A of [Steel \(1992\)](#) is that if all trees of a set of
 336 (binary) rooted 3-taxon trees R are displayed on a (binary) rooted N -taxon tree T
 337 and each internal edge of T is an internal edge of at least one tree in R , then T is
 338 the only N -taxon tree that displays all the 3-taxon trees of R . In other words, the
 339 N -taxon tree T can be identified from the set of 3-taxon trees R .

340 [Steel \(1992\)](#) note that an analogous theorem exists for unrooted quartets. Thus, the
 341 *unrooted* principal tree of the N -taxon CDM can be identified from the set of $\binom{N-1}{3}$
 342 topologies of *unrooted* 4-taxon principal trees that include the outgroup displayed on
 343 the *unrooted* principal tree of the N -taxon CDM. The principal tree of the N -taxon
 344 CDM is then rooted by the outgroup.

345 5A.1 Proof of Theorem 4

346 Finally, from Proposition 2A and Theorem 3A adapted to unrooted quartets that
347 include the outgroup, we can prove Theorem 4.

348 **Theorem 4** *Suppose CDM $\mathcal{N}$ has topology of principal tree $\mathcal{T}$. Suppose the BIC is used*
349 *for model selection in step 2 of Algorithm 1. Suppose $\hat{\mathcal{T}}$ is the estimate of $\mathcal{T}$ inferred by*
350 *Algorithm 1. Then there exists some constant $c > 0$ such that if the largest convergence*
351 *parameter of $\mathcal{N}$ is less than c ,*

$$\lim_{n \rightarrow \infty} \mathbb{P}(\hat{\mathcal{T}} = \mathcal{T}) = 1.$$

352 *Proof* Suppose $\mathcal{N}$ has a displayed 4-taxon CDM $\mathcal{N}_4$ with topology of principal tree $\mathcal{T}_4 =$
353 $(o, (a, (b, c)))$. Then from the proof of Theorem 2, for $\mathcal{N}_4$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 = 0,$$

354 while for some 4-taxon CDM with topology of principal tree $\mathcal{T}'_4 \neq (o, (a, (b, c)))$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 > 0.$$

355 Suppose m_1 corresponds to the union of sets of possible realizable phylogenetic tensors
356 for CDMs 1 – 5 for the topology of principal tree $\mathcal{T}_4$. Suppose also that m_2 corresponds
357 to the union of sets of possible realizable phylogenetic tensors for CDMs 1 – 5 for any 4-
358 taxon topology of principal tree that is not $\mathcal{T}_4$. Then $m_1 \cap m_2 = \emptyset$. Suppose $\theta \notin \overline{m}_1, \overline{m}_2$.
359 Then if $c > 0$ is sufficiently small, since the functions for the phylogenetic tensor elements
360 are analytic, the phylogenetic tensor corresponding to the CDM generating parameter is in
361 a neighborhood that includes a subset of m_1 , but none of m_2 . By Proposition 2A, m_1 is
362 selected by the BIC asymptotically with probability 1.

363 Next, we prove the claim that the set of inferred topologies of 4-taxon principal trees
364 equals the set of topologies of the principal trees of the 4-taxon CDMs displayed on $\mathcal{N}$. Then
365 from the adaptation of Theorem 3A to unrooted quartets, the topology of the principal tree of

366 $\mathcal{N}$ is the only topology that displays all inferred 4-taxon principal trees. Thus, any consistent
 367 supertree inference method used in step 3 of Algorithm 1 infers the topology of the principal
 368 tree of $\mathcal{N}$ consistently and the proof is complete.

369 All that is left to prove is the claim that the probability of the set of inferred 4-taxon
 370 principal trees equalling the set of topologies of principal trees of 4-taxon CDMs displayed
 371 on $\mathcal{N}$ converges to 1.

372 Suppose A_i is the event where the topology of the i^{th} 4-taxon principal tree is inferred
 373 incorrectly, given some arbitrary order. Then, by Proposition 2A, there exists some sample
 374 size n such that for $n' > n$, $\mathbb{P}(A_i) < \epsilon_i$ for some arbitrarily small $\epsilon_i > 0$. Then by Boole's
 375 inequality,

$$\mathbb{P}\left(\bigcup_{i=1}^{\binom{N-1}{3}} A_i\right) \leq \sum_{i=1}^{\binom{N-1}{3}} \mathbb{P}(A_i) < \sum_{i=1}^{\binom{N-1}{3}} \epsilon_i,$$

376 an arbitrarily small positive quantity. Thus, the set of topologies of the inferred 4-taxon
 377 principal trees of step 2 of Algorithm 1 equals the set of topologies of the principal trees of
 378 the 4-taxon CDMs displayed on $\mathcal{N}$ with probability converging to 1.

379 □

380 6A Controlling overfitting the CDM

381 The criteria $x^{(a)}$ and $y^{(a)}$ of Algorithm 2 limit overfitting of convergence groups to the
 382 inferred CDM. Further control of overfitting is achieved with a multiple comparisons
 383 correction, favoring 4-taxon trees over non-tree 4-taxon CDMs. For a given 4-taxon
 384 set that includes the outgroup taxon, the model selection criterion values are first
 385 converted into weights, for example, AIC or BIC (Burnham and Anderson 2004). These
 386 weights are a “tree weight” determined from the AIC or BIC of the tree and “non-tree
 387 weights” determined from the AIC or BIC values of the other CDMs. Tree weights
 388 could then be multiplied by some positive constant $b \geq 1$ to achieve further control for
 389 overfitting. A multiple comparisons correction, such as the Holm-Bonferroni method
 390 (Holm 1979), could then be applied to the tree weights over all 4-taxon sets that

include the outgroup taxon, as if the weights were p-values. If the tree is “rejected”, then the non-tree CDM with the lowest AIC or BIC is selected.

7A Proof of Proposition 5

Proposition 5 *For convergence group $C = \{c_1, c_2\}$ on CDM $\mathcal{N}$, let $a \in c_1$ and $b \in c_2$. Let v be the MRCA node of a and b , X_v be the set of leaf taxa descending from v and $X_C = c_1 \cup c_2$. Then the expected proportion of converging quartets for $\{a, b\}$ is*

$$\frac{|X_v \setminus X_C|}{N - 3} = \frac{|X_v| - |X_C|}{N - 3},$$

where $|X_v|$ and $|X_C|$ are the cardinalities of sets X_v and X_C .

Proof To determine the expected proportions of converging quartets, suppose taxa a and b are converging. Then convergence between these taxa can only be inferred on 4-taxon CDMs with topology of principal tree $(o, (a, (b, c)))$ or $(o, (b, (a, c)))$, for some arbitrary taxon c . With no loss of generality, we assume that the topology of the principal tree of some 4-taxon CDM is $(o, (a, (b, c)))$. To determine the expected proportions, we must determine the number of 4-taxon CDMs displayed on $\mathcal{N}$, displaying both a and b where they appear as non-sisters.

We start with the rooted tree $(o, (a, b))$ and append taxon c and include a convergence group C . One edge corresponding to the convergence group C must be ancestral to a , while the other must be ancestral to b . Thus, for C to be a non-sister convergence group, the remaining taxon c must be placed on an edge directly descended from v , corresponding to a speciation event before the epoch C is in. Thus, c could be any of the $|X_v \setminus X_C| = |X_v| - |X_C|$ taxa out of the $N - 3$ possible taxa that are not o , a or b . $\square$

8A Proof of Proposition 7

Proposition 7 *An arbitrary pair of distinct convergence groups on CDM $\mathcal{N}$ share no pair of converging leaf taxa.*

413 *Proof* Suppose C_1 and C_2 are two distinct convergence groups on $\mathcal{N}$. By Assumption 5 of
 414 Section 3.2, there can be at most one convergence group in each epoch. Thus, C_1 is either
 415 in an epoch before or after C_2 . With no loss of generality, we assume that C_1 is in an epoch
 416 before C_2 .

417 In order to share at least one pair of converging taxa, C_2 must be nested in C_1 . How-
 418 ever, by Assumption 9 of Section 3.2, there can be no convergence groups nested in other
 419 convergence groups.

420

□

421 9A Proof of Proposition 8

422 We assume that the topology of the principal tree of $\mathcal{N}$ is known. However, we note
 423 that if it is not known, from Theorem 4 it can be inferred consistently.

424 **Proposition 8** *The set of all convergence groups on CDM $\mathcal{N}$ can be identified from the set*
 425 *of displayed 4-taxon CDMs after suppressing sister convergence groups.*

426 *Proof* The set of displayed 4-taxon CDMs after suppressing sister convergence groups defines
 427 a matrix of proportions of converging quartets. However, in general the set of all convergence
 428 groups on $\mathcal{N}$ cannot be identified from the matrix (see Figure 5). Instead, we can identify a
 429 set of possible sets of convergence groups on $\mathcal{N}$ that correspond to the matrix of proportions
 430 of converging quartets. Since the set of displayed 4-taxon CDMs after suppressing sister
 431 convergence groups is assumed known, for the remainder of the proof we can restrict to this
 432 set of sets of convergence groups. We must then prove that we can identify the specific set of
 433 all convergence groups of $\mathcal{N}$.

434 If $\mathcal{N}$ is a tree, then the set of displayed 4-taxon CDMs after suppressing sister convergence
 435 groups is a set of trees. Thus, by Corollary 6, the matrix of proportions of converging quartets
 436 is the zero matrix. Alternatively, if $\mathcal{N}$ is not a tree, then $\mathcal{N}$ must have at least one non-sister
 437 convergence group. Call one such non-sister convergence group $C = \{c_1, c_2\}$, with v the most

438 recent common ancestral node of c_1 and c_2 . Then by Proposition 5, the expected proportion
 439 of converging quartets for $a \in c_1$ and $b \in c_2$ is $\frac{|X_v| - |X_C|}{N-3}$, where X_v is the set of all taxa
 440 descending from v and $|X_C| = |c_1| + |c_2|$. By the definition of non-sister convergence groups,
 441 $|X_v| - |X_C| > 0$. Thus, the matrix of converging quartets is not the zero matrix. Thus, if $\mathcal{N}$
 442 is a tree, the set of convergence groups can be identified from the set of displayed 4-taxon
 443 CDMs after suppressing sister convergence groups via the matrix of converging quartets.

444 For the remainder of the proof, we can assume that $\mathcal{N}$ is not a tree. Then the set of non-
 445 sister convergence groups defines a set S of 4-taxon CDMs displayed on $\mathcal{N}$ with non-sister
 446 convergence groups after suppressing sister convergence groups — note that 4-taxon CDMs
 447 of S can have one or two non-sister convergence groups. Suppose similarly that S' is a set of
 448 4-taxon CDMs defined by a set of non-sister convergence groups not on $\mathcal{N}$ but with the same
 449 matrix of proportions of converging quartets as the set of non-sister convergence groups on
 450 $\mathcal{N}$. We must prove that there exists some 4-taxon CDM in S that is not in S' . Then we can
 451 identify the set of convergence groups on $\mathcal{N}$ from the set of 4-taxon CDMs.

452 We prove that there is some 4-taxon CDM in S that is not in S' . We first consider an
 453 arbitrary 4-taxon CDM $\mathcal{N}_4$ in S . Consider arbitrary leaf taxon pair $\{a, b\}$, where $a \in c_1$ and
 454 $b \in c_2$. Furthermore, assume $c \in X_v \setminus X_C$. Then with no loss of generality, we can assume
 455 the topology of the principal tree of $\mathcal{N}_4$ is $(o, (b, (a, c)))$.

456 Suppose that $C' = \{c'_1, c'_2\}$ is one such non-sister convergence group that defines S' ,
 457 with c'_1, c'_2, v', X'_v and $X_{C'}$ as in Proposition 5. Now consider 4-taxon CDM $\mathcal{N}'_4$, defined by
 458 C' and on leaf taxon set $\{o, a, b, c\}$, with topology of principal tree $(o, (b, (a, c)))$. Since we
 459 require a non-sister convergence group on $\mathcal{N}'_4$ where a and b are both converging, we must
 460 have either $a \in c'_1$ and $b \in c'_2$ or $a \in c'_2$ and $b \in c'_1$. With no loss of generality, we assume
 461 that $a \in c'_1$ and $b \in c'_2$. Then $c_1 \subseteq c'_1$ or $c_1 \supset c'_1$. Similarly, $c_2 \subseteq c'_2$ or $c_2 \supset c'_2$. Both v and v'
 462 are the MRCA of a and b . Thus $v' = v$.

463 Now assume that $X_{C'} = X_C$. Then $c'_1 = c_1$ and $c'_2 = c_2$ and in turn, $C' = C$. Thus, S' is
 464 defined by a set of convergence groups that includes C and the 4-taxon CDM is in S' . Thus,
 465 we can assume that $X_{C'} \neq X_C$ and we cannot have both $c'_1 = c_1$ and $c'_2 = c_2$. However,
 466 since the matrices of proportions of converging quartets must be the same for the two sets of

467 convergence groups, we must have

$$\frac{|X_v| - |X_C|}{N - 3} = \frac{|X_{v'}| - |X_{C'}|}{N - 3},$$

468 which simplifies to $|X_C| = |X_{C'}|$, since $v' = v$. Thus, either $c_1 \subset c'_1$ and $c_2 \supset c'_2$ or $c_1 \supset c'_1$
 469 and $c_2 \subset c'_2$. With no loss of generality, we assume that $c_1 \subset c'_1$ and $c_2 \supset c'_2$.

470 Then there exists some choice of c such that $c \in c'_1 \setminus c_1$. For $c_2 \supset c'_2$, there must similarly
 471 be some taxon $d \in c_2 \setminus c'_2$. Thus, we are assuming that $N \geq 5$ — the outgroup and taxa a ,
 472 b , c and d . Then $a, c \in c'_1$ and $b \in c'_2$. Thus, before suppressing sister convergence groups to
 473 form $\mathcal{N}'_4$, C' must correspond with a sister convergence group on the 4-taxon CDM on leaf
 474 taxa $\{o, a, b, c\}$ — see Figure 1A for a graphical depiction of C and C' . Then any other choice
 475 of convergence group that defines S' , say C'' , must satisfy $c''_1 \supset c_1$ and the claim follows.
 476 Finally, since we have assumed $N \geq 5$, we must also consider $N = 4$. For $N = 4$, it is clear
 477 from the identifiability and distinguishability of all CDMs with no sister convergence that
 478 the claim holds.

479 $\square$

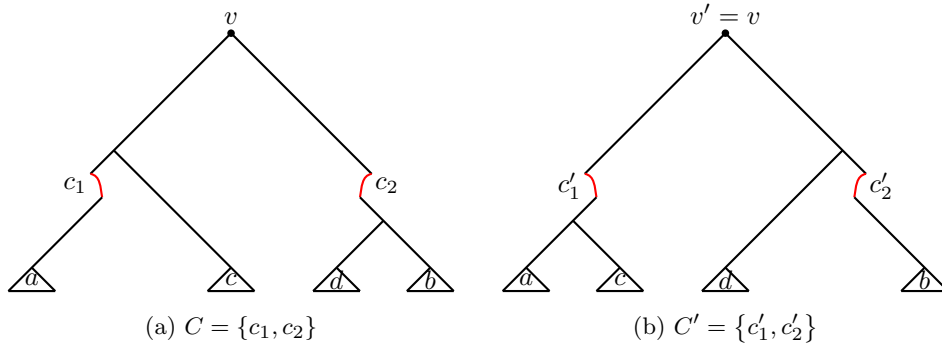


Fig. 1A Convergence groups C and C' . Labels c_1 , c_2 , c'_1 and c'_2 indicate the set of leaf taxa below that edge. Triangles are displayed CDMs. Labels inside triangles indicate one of possibly many taxa on leaves of those displayed CDMs. There may be more displayed CDMs not drawn below v that are not below either C or C'

480 10A Proof of Theorem 9

481 **Theorem 9** *Suppose CDM $\mathcal{N}$ has topology of principal tree $\mathcal{T}$ and convergence groups $\mathcal{G}$.
482 Suppose for all l , $\alpha_l = \beta_l$. Suppose for convergence group $\mathcal{C}_i = \{c_{1,i}, c_{2,i}\}$ that if $a \in c_{1,i} \cup c_{2,i}$,
483 then $a \notin c_{1,j} \cup c_{2,j}$ for any $j \neq i$. Suppose $\mathcal{T}$ is input into Algorithm 2, the BIC is used
484 for model selection in step 2, there are no multiple comparisons corrections and one such
485 tolerance criterion is $w = 1$. Suppose $\hat{\mathcal{G}}$ is the estimate of $\mathcal{G}$ inferred by Algorithm 2. Then
486 there exists some constant $c > 0$ such that if the largest convergence parameter of $\mathcal{N}$ is less
487 than c ,*

$$\lim_{n \rightarrow \infty} \mathbb{P}(\hat{\mathcal{G}} = \mathcal{G}) = 1.$$

488 *Proof* We start by finding expressions for the transformed phylogenetic tensors for various
489 4-taxon CDMs with and without sister convergence. We prove that the CDMs with sister
490 convergence are not distinguishable from the CDMs with the sister convergence groups sup-
491 pressed. Thus, regardless of whether the 4-taxon CDMs have sister convergence groups or
492 not, the non-sister convergence group is inferred consistently.

493 Since $\alpha_l = \beta_l$, $\gamma = 0$ and the transformed phylogenetic tensor for a 4-taxon CDM of
 494 Equation (1A) simplifies to

$$\hat{P} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ r_{0011} \\ 0 \\ r_{0101} \\ r_{0110} \\ 0 \\ 0 \\ r_{1001} \\ r_{1010} \\ 0 \\ r_{1100} \\ 0 \\ 0 \\ r_{1111} \end{bmatrix}.$$

495 With no loss of generality, we assume the topology of the principal tree of an arbitrary 4-
 496 taxon CDM displayed on $\mathcal{N}$ is $(o(a, (b, c)))$. Then of the possible convergence groups on the
 497 4-taxon CDM, the convergence group in the epoch closest to the root is the sister convergence
 498 group $C = \{\{a\}, \{b, c\}\}$. Thus, we first consider the distinguishability of two 4-taxon CDMs,
 499 one a tree and the other with this convergence group. For both CDMs we assume the tip
 500 epoch has epoch time 0. The first, which we call $\mathcal{N}_{4,1}$, is the tree $(o(a, (b, c)))$. Since the tip
 501 epoch has epoch time 0, taxa b and c are identical. The second CDM, which we call $\mathcal{N}_{4,2}$, has
 502 a single convergence group, $C = \{\{a\}, \{b, c\}\}$, followed by a speciation event involving b and
 503 c . Again, since the tip epoch has epoch time 0, taxa b and c are identical. See Figure 2A for
 504 a graphical depiction of the two CDMs. Suppose $\mathcal{N}_{4,1}$ has parameters with no apostrophes
 505 and $\mathcal{N}_{4,2}$ has parameters with apostrophes.

506 For $\mathcal{N}_{4,1}$ (see Mathematica file S12.nb (text version S13.txt) on [https://github.com/](https://github.com/jonathanmitchell88/CDMsSI)
 507 [jonathanmitchell88/CDMsSI](https://github.com/jonathanmitchell88/CDMsSI) for a derivation),

$$\left\{ \begin{array}{lcl} r_{0011} & = & 1, \\ r_{0101} & = & x_2 x_3, \\ r_{0110} & = & x_2 x_3, \\ r_{1001} & = & x_1 x_2, \\ r_{1010} & = & x_1 x_2, \\ r_{1100} & = & x_1 x_3, \\ r_{1111} & = & x_1 x_3. \end{array} \right.$$

508 For $\mathcal{N}_{4,2}$ (see Mathematica file S12.nb (text version S13.txt) for a derivation),

$$\left\{ \begin{array}{lcl} r_{0011} & = & 1, \\ r_{0101} & = & 1 - x'_4 (1 - x'_2 x'_3), \\ r_{0110} & = & 1 - x'_4 (1 - x'_2 x'_3), \\ r_{1001} & = & x'_1 x'_2 x'_4, \\ r_{1010} & = & x'_1 x'_2 x'_4, \\ r_{1100} & = & x'_1 x'_3 x'_4, \\ r_{1111} & = & x'_1 x'_3 x'_4. \end{array} \right.$$

509 For $\mathcal{N}_{4,1}$,

$$\left\{ \begin{array}{lcl} x_1 & = & \sqrt{\frac{r_{1001} r_{1100}}{r_{0101}}}, \\ x_2 & = & \sqrt{\frac{r_{0101} r_{1001}}{r_{1100}}}, \\ x_3 & = & \sqrt{\frac{r_{0101} r_{1100}}{r_{1001}}}, \\ r_{0101} & = & r_{0110}, \\ r_{1001} & = & r_{1010}, \\ r_{1100} & = & r_{1111}. \end{array} \right.$$

510 Since $x_1, x_2, x_3 \in (0, 1)$, for $\mathcal{N}_{4,1}$,

$$\left\{ \begin{array}{l} r_{0101} = r_{0110}, \\ r_{1001} = r_{1010}, \\ r_{1100} = r_{1111}, \\ r_{0101}r_{1001} < r_{1100}, \\ r_{0101}r_{1100} < r_{1001}, \\ r_{1001}r_{1100} < r_{0101}. \end{array} \right.$$

511 Similarly, for $\mathcal{N}_{4,2}$,

$$\left\{ \begin{array}{l} r_{0101} = r_{0110}, \\ r_{1001} = r_{1010}, \\ r_{1100} = r_{1111}. \end{array} \right.$$

512 Since we are assuming that all convergence parameters of $\mathcal{N}$ are less than some constant

513 $c > 0$, we can assume that $x'_4 = 1 - \epsilon$, where $\epsilon > 0$ is some small positive constant. Then for

514 $\mathcal{N}_{4,2}$,

$$\left\{ \begin{array}{l} r_{1100} - r_{0101}r_{1001} = x'_1x'_3(1 - x'^2_2) + O(\epsilon), \\ r_{1001} - r_{0101}r_{1100} = x'_1x'_2(1 - x'^2_3) + O(\epsilon), \\ r_{0101} - r_{1001}r_{1100} = x'_2x'_3(1 - x'^2_1) + O(\epsilon). \end{array} \right.$$

515 Since $c > 0$ can be chosen, there exists some choice of $\epsilon > 0$ sufficiently small such that
 516 for $\mathcal{N}_{4,2}$,

$$\left\{ \begin{array}{l} r_{0101} = r_{0110}, \\ r_{1001} = r_{1010}, \\ r_{1100} = r_{1111}, \\ r_{0101}r_{1001} < r_{1100}, \\ r_{0101}r_{1100} < r_{1001}, \\ r_{1001}r_{1100} < r_{0101}. \end{array} \right.$$

517 Thus, $\mathcal{N}_{4,1}$ and $\mathcal{N}_{4,2}$ are not distinguishable for this choice of $c > 0$. Thus, any 4-taxon
 518 CDM with $\alpha_l = \beta_l$ and this sister convergence group is not distinguishable from the CDM
 519 that results from suppressing the sister convergence. Thus, to determine the transformed
 520 phylogenetic tensor of any 4-taxon CDM with $\alpha_l = \beta_l$, we can assume there is no sister
 521 convergence in this epoch.

522 The next closest epoch to the root that could have a convergence group is the epoch
 523 just after taxa b and c have diverged. Thus, we compare the tree $(o, (a, (b, c)))$, which we
 524 call $\mathcal{N}_{4,3}$, and the CDM with topology of principal tree $(o, (a, (b, c)))$ and sister convergence
 525 group $\{\{b\}, \{c\}\}$ in the tip epoch, which we call $\mathcal{N}_{4,4}$. See Figure 3A for a graphical depiction
 526 of the two CDMs. Again, suppose $\mathcal{N}_{4,3}$ has parameters with no apostrophes and $\mathcal{N}_{4,4}$ has
 527 parameters with apostrophes.

528

For $\mathcal{N}_{4,3}$ (see Mathematica file S12.nb (text version S13.txt) for a derivation),

$$\left\{ \begin{array}{lcl} r_{0011} & = & x_4 x_5, \\ r_{0101} & = & x_2 x_3 x_4, \\ r_{0110} & = & x_2 x_3 x_5, \\ r_{1001} & = & x_1 x_2 x_4, \\ r_{1010} & = & x_1 x_2 x_5, \\ r_{1100} & = & x_1 x_3, \\ r_{1111} & = & x_1 x_3 x_4 x_5. \end{array} \right.$$

529

For $\mathcal{N}_{4,4}$ (see Mathematica file S12.nb (text version S13.txt) for a derivation),

$$\left\{ \begin{array}{lcl} r_{0011} & = & 1 - x'_6 (1 - x'_4 x'_5), \\ r_{0101} & = & x'_2 x'_3 x'_4 x'_6, \\ r_{0110} & = & x'_2 x'_3 x'_5 x'_6, \\ r_{1001} & = & x'_1 x'_2 x'_4 x'_6, \\ r_{1010} & = & x'_1 x'_2 x'_5 x'_6, \\ r_{1100} & = & x'_1 x'_3, \\ r_{1111} & = & x'_1 x'_3 (1 - x'_6 (x'_4 x'_5)). \end{array} \right.$$

530

For $\mathcal{N}_{4,3}$,

$$\left\{ \begin{array}{lcl} x_1 & = & \sqrt{\frac{r_{1001} r_{1100}}{r_{0101}}}, \\ x_2 & = & \sqrt{\frac{r_{0110} r_{1001}}{r_{0011} r_{1100}}}, \\ x_3 & = & \sqrt{\frac{r_{0101} r_{1100}}{r_{1001}}}, \\ x_4 & = & \sqrt{\frac{r_{0011} r_{0101}}{r_{0110}}}, \\ x_5 & = & \sqrt{\frac{r_{0011} r_{0110}}{r_{0101}}}, \\ r_{0101} r_{1010} & = & r_{0110} r_{1001}, \\ r_{0011} r_{1100} & = & r_{1111}. \end{array} \right.$$

531 Since $x_1, x_2, x_3, x_4, x_5 \in (0, 1)$, for $\mathcal{N}_{4,3}$,

$$\left\{ \begin{array}{l} r_{0101}r_{1010} = r_{0110}r_{1001}, \\ r_{0011}r_{1100} = r_{1111}, \\ r_{0011}r_{0101} < r_{0110}, \\ r_{0011}r_{0110} < r_{0101}, \\ r_{0101}r_{1100} < r_{1001}, \\ r_{0110}r_{1001} < r_{0011}r_{1100}, \\ r_{1001}r_{1100} < r_{0101}. \end{array} \right.$$

532 Similarly, for $\mathcal{N}_{4,4}$,

$$\left\{ \begin{array}{l} r_{0101}r_{1010} = r_{0110}r_{1001}, \\ r_{0011}r_{1100} = r_{1111}. \end{array} \right.$$

533 Since we are assuming that all convergence parameters of $\mathcal{N}$ are less than some constant

534 $c > 0$, we can assume that $x'_6 = 1 - \epsilon$, where $\epsilon > 0$ is some small positive constant. Then for

535 $\mathcal{N}_{4,4}$,

$$\left\{ \begin{array}{l} r_{0110} - r_{0011}r_{0101} = x'_2x'_3x'_5(1 - x'^2_4) + O(\epsilon), \\ r_{0101} - r_{0011}r_{0110} = x'_2x'_3x'_4(1 - x'^2_5) + O(\epsilon), \\ r_{1001} - r_{0101}r_{1100} = x'_1x'_2x'_4(1 - x'^2_3) + O(\epsilon), \\ r_{0011}r_{1100} - r_{0110}r_{1001} = x'_1x'_3x'_4x'_5(1 - x'^2_2) + O(\epsilon), \\ r_{0101} - r_{1001}r_{1100} = x'_2x'_3x'_4(1 - x'^2_1) + O(\epsilon). \end{array} \right.$$

536 Since $c > 0$ can be chosen, there exists some choice of $\epsilon > 0$ sufficiently small such that
 537 for $\mathcal{N}_{4,4}$,

$$\left\{ \begin{array}{l} r_{0101}r_{1010} = r_{0110}r_{1001}, \\ r_{0011}r_{1100} = r_{1111}, \\ r_{0011}r_{0101} < r_{0110}, \\ r_{0011}r_{0110} < r_{0101}, \\ r_{0101}r_{1100} < r_{1001}, \\ r_{0110}r_{1001} < r_{0011}r_{1100}, \\ r_{1001}r_{1100} < r_{0101}. \end{array} \right.$$

538 Thus, $\mathcal{N}_{4,3}$ and $\mathcal{N}_{4,4}$ are not distinguishable for this choice of $c > 0$. Thus, any 4-taxon
 539 CDM with $\alpha_l = \beta_l$ and this sister convergence group is not distinguishable from the CDM
 540 that results from suppressing the sister convergence. Thus, to determine the transformed
 541 phylogenetic tensor of any 4-taxon CDM with $\alpha_l = \beta_l$, we can again assume there is no sister
 542 convergence in this epoch.

543 By the assumption that no leaf taxa belong to more than one convergence group, there
 544 can be no more than one convergence group on any arbitrary 4-taxon CDM displayed on
 545 $\mathcal{N}$. Thus, taking into consideration $\mathcal{N}_{4,1}$ and $\mathcal{N}_{4,2}$ not being distinguishable and $\mathcal{N}_{4,3}$ and
 546 $\mathcal{N}_{4,4}$ not being distinguishable, we can conclude that any arbitrary 4-taxon CDM displayed
 547 on $\mathcal{N}$ is not distinguishable from the 4-taxon CDM that results from suppressing any sister
 548 convergence group, which is one of CDM 1 – 3 of Figure 2.

549 Next, we establish that CDM 3 is identifiable under these assumptions. For this CDM,
 550 which we call $\mathcal{N}_{4,5}$ (see Mathematica file S12.nb (text version S13.txt)),

$$\left\{ \begin{array}{l} r_{0011} = x_4 x_5 x_6 x_7, \\ r_{0101} = x_2 x_3 x_4 x_6 x_8, \\ r_{0110} = x_7 x_8 (1 - x_6 (1 - x_2 x_3 x_5)), \\ r_{1001} = x_1 x_2 x_4, \\ r_{1010} = x_1 x_2 x_5 x_6 x_7, \\ r_{1100} = x_1 x_3 x_6 x_8, \\ r_{1111} = x_1 x_4 x_7 x_8 (x_2 (1 - x_6) + x_3 x_5 x_6). \end{array} \right.$$

551 In terms of the set of parameters $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$ of Section 2A.1,

$$\left\{ \begin{array}{l} r_{0011} = y_4 y_5 y_6, \\ r_{0101} = y_2 y_3 y_4 y_6, \\ r_{0110} = y_7 (1 - y_6) + y_2 y_3 y_5 y_6, \\ r_{1001} = y_1 y_2 y_4, \\ r_{1010} = y_1 y_2 y_5 y_6, \\ r_{1100} = y_1 y_3 y_6, \\ r_{1111} = y_1 (y_2 y_4 y_7 (1 - y_6) + y_3 y_5 y_6). \end{array} \right.$$

552 In S14.m2 (output file S15.txt) on <https://github.com/jonathanmitchell88/CDMsSI>, we
 553 see that the set of parameters $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7\}$ is identifiable. It follows that CDMs
 554 1 and 2 are also identifiable.

555 Thus, using similar arguments to those of the proof of Theorem 4, with probability
 556 converging to 1, step 2 of Algorithm 2 infers all the 4-taxon CDMs with the outgroup that
 557 are displayed on $\mathcal{N}$ after suppressing sister convergence groups.

558 If $\mathcal{N}$ is a tree, then $s = 0$ in step 4 of Algorithm 2, the algorithm terminates and the
 559 tree is returned. If $\mathcal{N}$ is not a tree, since $w = 1$, a potential convergence group on $\mathcal{N}$ is only
 560 considered if, for all pairs of converging taxa in the convergence group, the inferred 4-taxon

561 CDMs with that pair of taxa as non-sisters all have the pair converging. Thus, asymptotically
 562 with probability 1, only convergence groups on $\mathcal{N}$ can be on the inferred N -taxon CDM. If
 563 not all convergence groups of $\mathcal{N}$ have been included on the inferred CDM, then there are
 564 some elements of $\mathbf{O}$ that are non-zero corresponding to elements of $\mathbf{E}$ that are zero. These
 565 elements correspond to the pairs of converging taxa in convergence groups of $\mathcal{N}$ that are not
 566 yet on the inferred CDM. Including these convergence groups on the inferred CDM makes
 567 these elements of $\mathbf{E}$ equal to the corresponding elements of $\mathbf{O}$, decreasing the sum of squared
 568 differences. Once all convergence groups of $\mathcal{N}$ have been appended to the inferred CDM,
 569 $\mathbf{O} = \mathbf{E}$. Thus, no more convergence groups can be appended to the inferred CDM to decrease
 570 the sum of squared differences and the algorithm terminates.

□

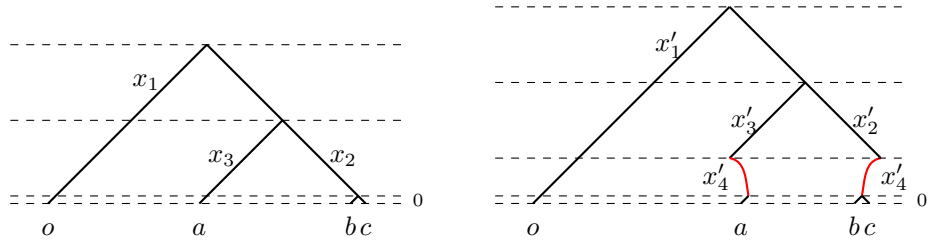


Fig. 2A Two CDMs that are not distinguishable under the assumptions of Theorem 9

11A Inferring convergence group orders on N -taxon CDMs

574 The next algorithms infer partial orders on the convergence groups and determine
 575 whether or not there is a convergence group in the tip epoch. CDMs 4 and 5 have two
 576 convergence groups and thus provide power to determine convergence group orders.
 577 Whether or not there is a convergence group in the tip epoch can also be determined

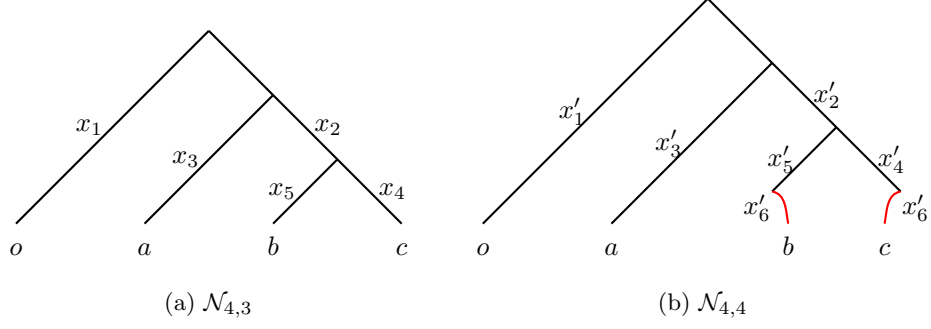


Fig. 3A Two CDMs that are not distinguishable under the assumptions of Theorem 9

578 from the inferred 4-taxon CDMs. For example, CDM 2 and CDM 3 differ by CDM 2
 579 having its convergence group in the tip epoch versus CDM 3 having its convergence
 580 group in the epoch before the tip epoch.

581 Suppose an inferred 4-taxon CDM has two non-sister convergence groups. One of
 582 the edges of the 4-taxon principal tree corresponds to a converging taxon in both
 583 convergence groups. The order of these convergence groups may not be determined by
 584 the matrix of edge partial orders from Algorithm 2. If the convergence group order is
 585 not determined, we determine which order is best supported by selecting CDMs from
 586 those with the appropriate convergence groups with a model selection procedure.

587 For convergence groups C_i and C_j , whose partial order has not been determined, we
 588 tally proportions of 4-taxon CDMs best supported by the two orders to obtain a matrix
 589 of “observed” convergence group order ratios. Convergence group orders are resolved
 590 in a stepwise fashion by minimizing the sum of squared differences between matrices
 591 of observed and “expected” partial convergence group order ratios. The matrix of
 592 inferred convergence group order ratios is updated after each convergence group order
 593 is inferred. We discard 4-taxon CDMs with convergence group orders not consistent
 594 with the matrix of inferred convergence group orders.

595 Suppose an arbitrary convergence group is $C_i = \{c_{1,i}, c_{2,i}\}$. On the N -taxon CDM,
 596 if $|c_{1,i}| > 1$ and/or $|c_{2,i}| > 1$ or C_i is in an epoch before another convergence group,

597 then C_i cannot be in the tip epoch. For other convergence groups, whether they are
598 in the tip epoch or not must be inferred.

599 For each 4-taxon CDM with a fixed leaf labeling with a possible convergence group
600 in the tip epoch, we determine which CDM is best supported among the two CDMs,
601 for example, CDM 2 versus CDM 3 or CDM 4 versus CDM 5. For C_i , we tally the 4-
602 taxon CDMs displaying the given convergence group with and without the convergence
603 group in the tip epoch.

604 If C_i corresponds to a greater proportion of 4-taxon CDMs with the convergence
605 group in the tip epoch than any other convergence group and the proportion is greater
606 than some cutoff, for example, half, then we infer that C_i is in the tip epoch. We
607 retain only one possible CDM for each 4-taxon set after the convergence group order
608 has been assigned and it has been determined which, if any, convergence group is in
609 the tip epoch.

610 Note that some convergence group orders may still be undefined. Suppose two
611 convergence groups do not have an order defined by the edge partial order of the
612 principal tree or the orders of other convergence groups. Suppose both convergence
613 groups are only ever present on 4-taxon CDMs where one convergence group is a sister
614 convergence group. Then there will be no information to resolve the order of these two
615 convergence groups. We leave these convergence group orders unresolved. Thus, we
616 have a *partial* order on the convergence groups. Algorithms 1A and 2A for inferring
617 convergence group orders and any convergence group in the tip epoch then follow.

618 We do not prove consistency of inference of the convergence group partial orders
619 from Algorithm 1A. This is because Theorem 9 assumes that no leaf taxa belong to
620 more than one convergence group. Thus, all 4-taxon CDMs displayed on $\mathcal{N}$ have at
621 most one non-sister convergence group and there are no convergence group orders to
622 infer. Furthermore, we do not prove consistency of inference of the convergence groups
623 in the tip epoch.

Algorithm 1A Convergence group order inference

Input: N -taxon CDM $\hat{\mathcal{N}}$ comprising N -taxon topology of principal tree $\hat{\mathcal{T}}$ and list of convergence groups $\hat{\mathcal{G}}$, as well as $\binom{N-1}{3} \times 27$ matrix of model selection criterion values $\mathbf{M}$ and matrix of partial edge orders $\mathbf{P}$.

1. Initialize empty list of inferred 4-taxon CDMs L_Q . Initialize $k \times k$ matrix of observed convergence group orders $\mathbf{O}$ as zero matrix, where k is length of list $\hat{\mathcal{G}}$. Initialize $k \times k$ matrix $\mathbf{E}$ of expected convergence group orders as convergence group orders defined by $\mathbf{P}$, with $[\mathbf{E}]_{ij} = 1$ if convergence group i before j and 0 otherwise.
2. For each 4-taxon set that includes outgroup o , with model selection criterion, select CDM from those displayed on $\hat{\mathcal{N}}$ and permitted by $\mathbf{E}$ and append to L_Q .
3. For all i, j , compute $[\mathbf{O}]_{ij}$ as proportion of inferred 4-taxon CDMs displaying convergence groups i and j , where i is before j .
4. Compute initial sum of squared differences between elements of $\mathbf{O}$ and $\mathbf{E}$, $s = \sum_{i=1}^k \sum_{j=1}^k ([\mathbf{O}]_{ij} - [\mathbf{E}]_{ij})^2$.
5. Assign new order between two convergence groups that minimizes s .
6. Update $\mathbf{E}$ and s to reflect newly inferred convergence group order. Suppose new order is convergence group x before y . Then all convergence groups above x are also above y and all convergence groups below y are also below x . If no pairs of convergence groups left to assign orders to, terminate algorithm.
7. Return to Step 5.

Output: N -taxon CDM $\hat{\mathcal{N}}$ comprising N -taxon topology principal tree $\hat{\mathcal{T}}$ and list of convergence groups $\hat{\mathcal{G}}$, as well as $\binom{N-1}{3} \times 27$ matrix of model selection criterion values $\mathbf{M}$, matrix of partial edge orders $\mathbf{P}$ and matrix of expected convergence group orders $\mathbf{E}$.

624 However, if all inferred 4-taxon CDMs that include the outgroup are the 4-taxon
625 CDMs displayed on the generating N -taxon CDM after suppressing sister convergence
626 groups, then it is straightforward to prove that Algorithm 1A correctly infers all
627 orders of convergence groups of the generating N -taxon CDM that can be determined
628 from the displayed 4-taxon CDMs. Furthermore, it is also straightforward to prove
629 that Algorithm 2A correctly infers which, if any, convergence group of the generating
630 N -taxon CDM is in the tip epoch.

631 12A Proof of Proposition 10

632 **Proposition 10** *All edge lengths of the principal tree of each of CDM 1 – 5 are identifiable.*

Algorithm 2A Inference of convergence groups in tip epochs

Input: N -taxon CDM $\hat{\mathcal{N}}$ comprising N -taxon topology principal tree $\hat{\mathcal{T}}$ and list of convergence groups $\hat{\mathcal{G}}$, as well as $\binom{N-1}{3} \times 27$ matrix of model selection criterion values $\mathbf{M}$, matrix of partial edge orders $\mathbf{P}$, matrix of expected convergence group orders $\mathbf{E}$ and tolerance $\tau \in [0, 1]$.

1. Initialize empty list of inferred 4-taxon CDMs L_Q . Initialize vector $\mathbf{D}$ of length k of convergence groups in tip epoch as zero vector, where k is length of list $\hat{\mathcal{G}}$.
2. For each 4-taxon set that includes outgroup o , select CDM from those displayed on $\hat{\mathcal{N}}$ and permitted by $\mathbf{E}$ with model selection criterion and append to L_Q .
3. For all i , if convergence group $C_i = \{c_{1,i}, c_{2,i}\}$ satisfies $|c_{1,i}| = |c_{2,i}| = 1$ and is not before any other convergence group of $\hat{\mathcal{N}}$, compute $[\mathbf{D}]_i$ as proportion of inferred 4-taxon CDMs with C_i in tip epoch.
4. If $\max_{i \in \{1, 2, \dots, k\}} [\mathbf{D}]_i = [\mathbf{D}]_j$ and $D_j > \tau$, set $[\mathbf{D}]_j = 1$.

Output: N -taxon CDM $\hat{\mathcal{N}}$ comprising N -taxon topology principal tree $\hat{\mathcal{T}}$ and list of convergence groups $\hat{\mathcal{G}}$, as well as $\binom{N-1}{3} \times 27$ matrix of model selection criterion values $\mathbf{M}$, matrix of partial edge orders $\mathbf{P}$, matrix of expected convergence group orders $\mathbf{E}$ and vector of convergence groups in tip epoch $\mathbf{D}$.

633 *Proof* Using the parameterization of Section 2A.1, for CDM 5, with principal tree

634 $(o, (a, (b, c)))$, the sums of edge lengths between leaf taxa are

$$\left\{ \begin{array}{l} d_{o,a} = l_1 + l_3 + l_6 + l_8 + l_9 + l_{11} = -\log(x_1 x_3 x_6 x_8 x_9 x_{11}) = -\log(y_1 y_3 y_6 y_8), \\ d_{o,b} = l_1 + l_2 + l_5 + l_6 + l_7 = -\log(x_1 x_2 x_5 x_6 x_7) = -\log(y_1 y_2 y_5 y_6), \\ d_{o,c} = l_1 + l_2 + l_4 + l_9 + l_{10} = -\log(x_1 x_2 x_4 x_9 x_{10}) = -\log(y_1 y_2 y_4 y_8), \\ d_{a,b} = l_2 + l_3 + l_5 + 2l_6 + l_7 + l_8 + l_9 + l_{11} = -\log(x_2 x_3 x_5 x_6^2 x_7 x_8 x_9 x_{11}) \\ \quad = -\log(y_2 y_3 y_5 y_6^2 y_8), \\ d_{a,c} = l_2 + l_3 + l_4 + l_6 + l_8 + 2l_9 + l_{10} + l_{11} = -\log(x_2 x_3 x_4 x_6 x_8 x_9^2 x_{10} x_{11}) \\ \quad = -\log(y_2 y_3 y_4 y_6 y_8^2), \\ d_{b,c} = l_4 + l_5 + l_6 + l_7 + l_9 + l_{10} = -\log(x_4 x_5 x_6 x_7 x_9 x_{10}) = -\log(y_4 y_5 y_6 y_8). \end{array} \right.$$

635 From Equations (3A), the set $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$ is identifiable. Thus, the set

636 $\{d_{o,a}, d_{o,b}, d_{o,c}, d_{a,b}, d_{a,c}, d_{b,c}\}$ is also identifiable for CDM 5. Solving for the lengths of the

edges of the principal tree,

$$\begin{cases} l_o = \frac{1}{2} (d_{o,a} + d_{o,b} - d_{a,b}), \\ l_a = \frac{1}{2} (d_{o,a} - d_{o,b} + d_{a,b}), \\ l_b = \frac{1}{2} (d_{a,b} - d_{a,c} + d_{b,c}), \\ l_c = \frac{1}{2} (-d_{a,b} + d_{a,c} + d_{b,c}), \\ l_{bc} = \frac{1}{2} (-d_{o,a} + d_{o,b} + d_{a,c} - d_{b,c}), \end{cases}$$

where l_o is the sum of divergence parameters along the two edges of the principal tree whose parent node is the root, l_a , l_b and l_c are the sums of divergence and possibly convergence parameters along the terminal edges whose descendent leaf taxa are a , b and c respectively and l_{bc} is the sum of divergence parameters along the edge whose descendent leaf taxa are b and c .

It follows that all edge lengths are also identifiable for CDMs 1 – 4 since expressions for the sums of edge lengths are the same, except that some $y_i = 1$.

□

13A Proof of Proposition 11

Proposition 11 *All convergence parameters of each of CDM 2 – 5 are identifiable.*

Proof On CDM 5, parameters $y_6 = x_6$ and $y_8 = x_9$ are identifiable. Thus, the convergence parameters $l_6 = a_6 + b_6 = -\log(y_6)$ and $l_9 = a_9 + b_9 = -\log(y_8)$ are identifiable. Thus, for all other CDMs with these convergence parameters, they are also identifiable. □

14A Proof of Proposition 12

Proposition 12 *The root parameter $\gamma = [\Pi]_0 - [\Pi]_1$, where $[\Pi]_0$ and $[\Pi]_1$ are the probabilities of states 0 and 1 at the root, respectively, is identifiable on each of CDM 1 – 5.*

654 *Proof* From Equation (1A) for the phylogenetic tensor of CDM 5, $q_{0001} = q_{0010} = q_{0100} =$
655 $q_{1000} = \gamma$. Thus, γ is identifiable for CDM 5. Since all other CDMs are nested in CDM 5 and
656 none correspond to generic values of γ — instead they correspond to some generic values of
657 x_i or y_i — γ is also identifiable for CDMs 1 – 4. $\square$

658 15A Proof of Theorem 13

659 **Theorem 13** *Suppose CDM $\mathcal{N}$ has topology of principal tree $\mathcal{T}$, convergence groups $\mathcal{G}$, prin-*
660 *cipal tree edge lengths $\mathbf{l}$, root parameter γ and convergence parameters $\mathbf{v}$. Suppose $\mathcal{T}$, $\mathcal{G}$,*
661 *convergence group partial orders and tip epoch convergence groups of $\mathcal{N}$ are input into Algo-*
662 *rithm 3. Suppose in step 4 of Algorithm 3 only 4-taxon sets for which 4-taxon CDMs displayed*
663 *on $\mathcal{N}$ have no sister convergence are considered. Suppose that for each convergence group of*
664 *$\mathcal{G}$ — say $C_a = \{c_{1,a}, c_{2,a}\}$ — there is at least one 4-taxon CDM displayed on $\mathcal{N}$ with no*
665 *sister convergence where $x \in c_{1,a}$, $y \in c_{2,a}$ are non-sister leaf taxa on the displayed CDM.*
666 *Suppose further that matrix $\mathbf{X}$ in step 6 of Algorithm 3 has rank $2N - 3$. Suppose $\hat{\mathbf{l}}$, $\hat{\gamma}$ and*
667 *$\hat{\mathbf{v}}$ are the estimates of $\mathbf{l}$, γ and $\mathbf{v}$, respectively, inferred by Algorithm 3. Then for any $\epsilon > 0$,*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \hat{\mathbf{l}} - \mathbf{l} \right| > \epsilon \right) = 0, \quad \lim_{n \rightarrow \infty} \mathbb{P} (|\hat{\gamma} - \gamma| > \epsilon) = 0, \quad \lim_{n \rightarrow \infty} \mathbb{P} (|\hat{\mathbf{v}} - \mathbf{v}| > \epsilon) = 0,$$

668 where $\left| \hat{\mathbf{l}} - \mathbf{l} \right|$ and $|\hat{\mathbf{v}} - \mathbf{v}|$ involve l^1 norms.

669 *Proof* In step 4 of Algorithm 3, only 4-taxon sets that include the outgroup for which 4-taxon
670 CDMs displayed on $\mathcal{N}$ have no sister convergence are considered. Thus, all such 4-taxon
671 CDMs displayed on $\mathcal{N}$ are CDM 1 – 5. Since some 4-taxon sets may not be considered, we
672 cannot yet assume that all parameters are identifiable. However, for a given 4-taxon set that
673 is considered, from the proof of Proposition 10, all sums of edge lengths between leaf taxa in
674 the 4-taxon set are identifiable. From Propositions 11 and 12, all convergence parameters on
675 the 4-taxon CDM displayed on $\mathcal{N}$ and the root parameter γ are also identifiable.

676 Thus, for the given 4-taxon set, the estimates of sums of edge lengths between taxa formed
677 from the sums of maximum likelihood estimates of parameters converge in probability to the
678 sums of edge lengths between taxa for $\mathcal{N}$. Likewise, the maximum likelihood estimates of the

679 convergence parameters converge in probability to the convergence parameters on $\mathcal{N}$ and the
680 maximum likelihood estimate of γ also converges in probability to γ . Thus, it follows that
681 when averaging over all 4-taxon sets that are considered, the estimates of the sums of edge
682 lengths between taxa converge in probability to the values for $\mathcal{N}$.

683 Now, since the matrix $\mathbf{X}$ has rank $2N - 3$, $\mathbf{X}^T \mathbf{X}$ is invertible. It follows that $\hat{\mathbf{l}}$ also
684 converges in probability to $\mathbf{l}$ in step 7 of Algorithm 3. By assumption, for each convergence
685 group of $\mathcal{G}$ there is at least one 4-taxon CDM displayed on $\mathcal{N}$ where two converging taxa
686 of the convergence group are non-sister taxa and there is no sister convergence. Thus, each
687 convergence parameter of $\hat{\mathbf{v}}$ is estimated at least once. Thus, $\hat{\mathbf{v}}$ converges in probability to $\mathbf{v}$.
688 Finally, since γ is fixed across all 4-taxon CDMs displayed on $\mathcal{N}$, to be consistently estimated
689 it only needs to be estimated for one 4-taxon CDM displayed on $\mathcal{N}$. Thus, $\hat{\gamma}$ converges in
690 probability to γ .

691 $\square$

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